

When does adapting a queue discipline’s parameters pay? A scaling law in target/RTT, measured on stock Linux

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Abstract

Linux ships `fq_codel` as its default queue discipline with four parameters fixed at configuration time. Whether adapting those parameters while traffic runs is worth doing has been answered both ways in the literature, and neither answer states the conditions under which the other holds. Here we measure 210 runs spanning three workloads, eight alternative queue disciplines and path round-trip times from 2 to 200 ms, and find that the benefit of adaptation is governed not by path RTT but by the dimensionless ratio $r = \text{target}/\text{RTT}$. Benefit follows a logistic curve in r with a ceiling of 20.5 % and half of that reached at $r = 0.50$ ($R^2 = 0.986$), a form that holds across a 16-fold change in RTT and transfers to two further AQM mechanisms. Published configuration guidance places a correctly configured deployment at r between 0.05 and 0.10, so half the available benefit lies five to ten times outside that envelope, which makes parameter adaptation a remedy for misconfiguration rather than an improvement on correct configuration. Two measured results frame this. Flow queueing combined with a delay target reduces tail latency by 99.0 % relative to an unmanaged queue at indistinguishable goodput, an effect two orders of magnitude larger than any parameter tuning recorded here; and CAKE, an existing discipline requiring no controller, outperforms the adapted system at every path RTT tested. All hypotheses and their decision thresholds were registered in a public repository before the data existed.

Keywords: active queue management; bufferbloat; `fq_codel`; CoDel; dimensionless scaling law; runtime parameter adaptation; Linux traffic control; pre-registered evaluation

1 Introduction

Bufferbloat is the inflation of queueing delay by buffers larger than any control loop requires, with throughput left unaffected[1]. The Linux response is `fq_codel`, which applies Deficit Round Robin across roughly a thousand flow queues and runs an independent CoDel instance in each[2]. Four parameters govern it and none changes once set: `target` (5 ms by default), `interval` (100 ms), `limit` and `quantum`.

Adapting those parameters while traffic runs has obvious appeal and clear precedent. Adaptive RED adjusts a maximum drop probability by an additive-increase multiplicative-decrease rule[3]; Ye and Leung derive stability conditions for CoDel and adapt its `interval` analytically[4]; DESiRED adapts an AQM’s delay target in P4 using deep reinforcement learning and in-band telemetry[5]. Runtime adaptation of a delay target is therefore not novel, and no claim of novelty is made for it here.

Three families of queue discipline bear on this. Random Early Detection opened the field by dropping with a probability derived from average queue length[6], and its sensitivity to configuration is what prompted Adaptive RED. PIE replaced queue length with a delay estimate inside a proportional-integral controller[7]. CoDel abandoned queue length altogether, dropping when the minimum sojourn time over a sliding interval exceeds a target[8, 9]; its authors argue

that the two parameters are expressed relative to path round-trip time by construction and therefore need no tuning, and Equation 1 below quantifies the point at which that argument stops holding. Beyond `fq_code1`, CAKE integrates shaping, flow queueing and host fairness in one discipline[10], FQ-PIE is the corresponding combination for PIE[11], and L4S standardises a low-latency service class built on a dual-queue coupled AQM[12, 13]. All of these except L4S, whose dual-queue structure needs endpoint support this testbed does not provide, are measured here against one identical bottleneck, alongside an unmanaged queue and SFQ.

Among systems that adapt at runtime, QueuePilot learns a marking policy for small buffers by reinforcement learning[14] and AQM-LLM distils a language model into a marking controller[15]. Toopchinezhad and Ahmadi survey the machine-learning AQM literature and observe that heuristic schemes require careful parameter adjustment, which limits their applicability in deployment[16]. SCRr revisits fair-queueing scheduling for modern link rates[17], and Ray et al. characterise how the presence of an AQM distorts speed-test measurement[18]. Table 1 places the present work against this line. Each of these systems reports that its adaptation improved something, and each adapts a single quantity; none states the conditions under which that adaptation is worth running, which is the gap addressed here.

Table 1 The present work against six recent systems. Each adapts at most a single quantity and reports a gain for doing so; DualPI2 is included as the standardised alternative that adapts nothing. None states the condition under which adaptation is worth running.

System	Year	Quantity adapted	Method
Adaptive RED[3]	2001	maximum drop probability	AIMD
ACoDel[4]	2020	interval	analytical
QueuePilot[14]	2023	marking probability	offline RL
DualPI2 / L4S[13]	2023	none	dual queue
DESiRED[5]	2024	target delay	deep RL
AQM-LLM[15]	2025	action policy	distilled LLM
This work	2026	four parameters	AIMD with gradient

What that body of work does not supply is the condition under which adaptation pays. Systems report gains, independent evaluations report none, and neither camp states where the boundary between the two lies. RFC 7567 meanwhile recommends active queue management that requires no per-flow configuration[19], which makes a null result for adaptation a reportable outcome rather than a failed experiment. Two pieces of apparatus are missing throughout as well. A controller that merely occupies the processor is hard to distinguish from one that helps unless a third arm runs the same code and applies nothing; and the sparse probe traffic used to measure latency is usually averaged into the bulk flows, even where the discipline under test was designed to treat those two populations differently.

This study supplies the missing condition and the apparatus needed to establish it. The condition turns out to be a dimensionless ratio rather than any absolute quantity, which is what allows it to be stated once and applied to deployments of different speed and distance.

2 Results

Every value reported here is generated from the committed measurement logs by the analysis scripts; none is entered by hand. Unless stated otherwise, figures describe the steady workload at a 20 ms base path RTT over a 10 Mbit/s bottleneck, with means taken over three independent runs and uncertainties given as 95 % confidence intervals.

2.1 Comparison of the four measured outputs across all systems

Figure 1 sets out the four outputs the study measures, side by side, for all ten configurations. Tail latency of the sparse probe and mean latency of the bulk flows are reported separately

throughout, because flow-queueing disciplines privilege sparse flows by construction and a single averaged latency conceals that. Queue occupancy is the mechanism those latencies arise from, and goodput is the quantity a latency improvement might have been bought with.

The dominant effect is the presence of flow queueing with a delay target, not the tuning of one. An unmanaged first-in-first-out queue reaches 2344ms at the 95th percentile against 24.57ms for static `fq_codel`, a 99.0 % reduction ($p < 0.001$), while goodput is statistically indistinguishable between them (9.453 against 9.414 Mbit/s, $p = 0.20$). The goodput panel is drawn from zero precisely so that this can be seen: every system delivers the same throughput, so the latency ordering across three orders of magnitude is a property of queue management and not a rate trade.

Fair queueing on its own is not sufficient. SFQ isolates flows but applies no delay target, and its bulk queues sit at their 127-packet limit throughout, so its sparse probe reads 30.68 ms while the bulk flows sharing the same link read 294.8 ms. The scheduler and the AQM perform distinct jobs and both are required.

The three `fq_codel` arms, static, sham and adapted, are not separable from one another at this scale on any of the four outputs, which is the first indication that parameter tuning is a second-order effect.

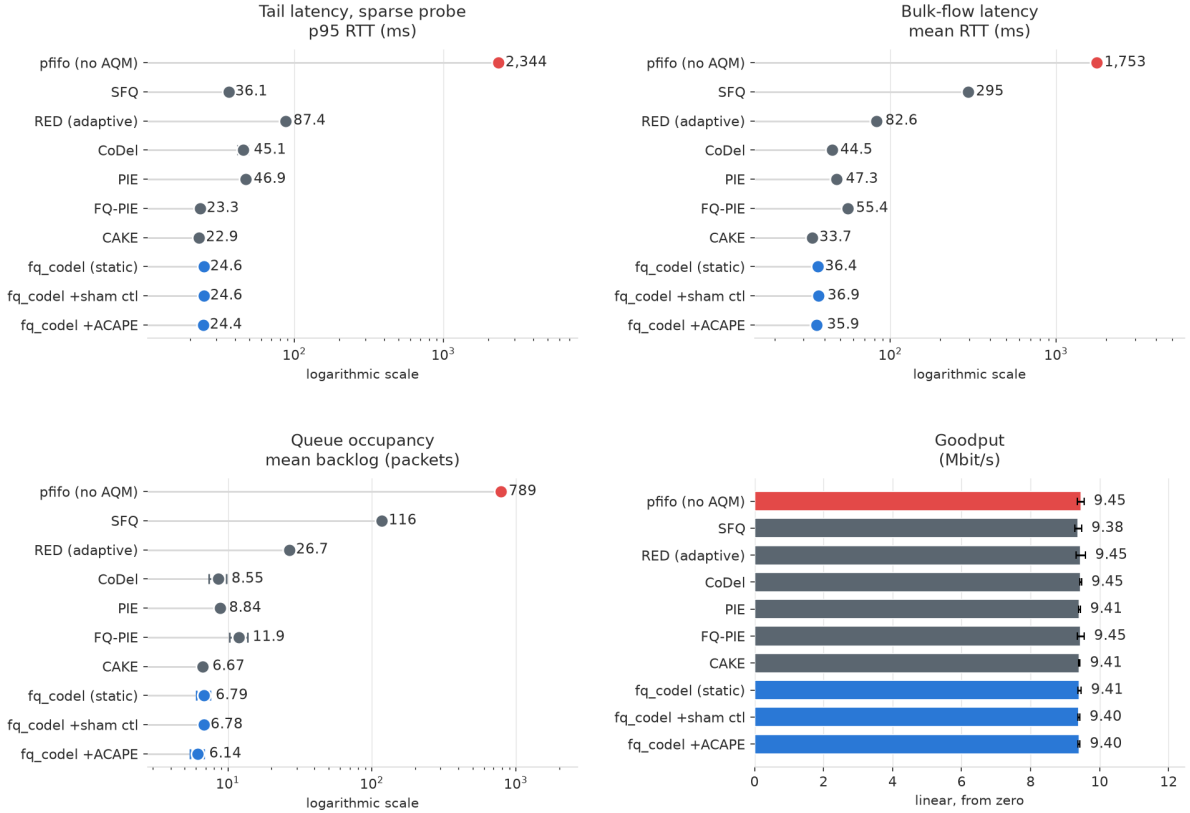


Fig. 1 The four measured outputs compared across all ten configurations under the steady workload. Lower is better for the three latency and occupancy panels, higher is better for goodput. Red marks the unmanaged baseline, blue the three `fq_codel` arms including the adapted system. The first three outputs span orders of magnitude and are drawn on a logarithmic axis, where position carries the value; goodput is linear from zero, where bar length carries it. Points and bars are means over three runs and whiskers are 95 % confidence intervals.

2.2 Effect of adaptation, parameter by parameter

Figure 2 gives every measured quantity against every system in one view, with each row’s rank shown as colour and each row’s true range printed beside it. The rank encoding makes within-row ordering legible across quantities whose units differ by orders of magnitude; the range column is there so that a row with strong colour contrast and no meaningful difference, such as goodput or fairness at $1.01\times$, is not read as a result.

Under the steady workload the largest movement the controller produces is in mean queue occupancy, from 6.79 to 6.14 packets. That is a 9.6 % reduction as a point estimate, but its 95 % confidence interval spans ± 1.00 packets and so covers zero, and the same is true of the other three outputs. The sham arm narrows what the null can be attributed to: the static and sham arms also lie within each other’s intervals on all four outputs, so the controller’s computational cost is not detectable either, and the absence of an effect is not the cost of running the loop masking a benefit. Under the staged workload, built specifically to exercise adaptation by varying the flow count from two to twenty-four during the run, nothing separates there either, and the adjustments are seen to lag the transitions they respond to. Section 3 returns to why a null result is expected at this operating point.

The comparison that matters for practice is with CAKE, which requires no controller, no telemetry and no model of the queue. It reaches 22.9 ms at the 95th percentile against 24.43 ms for the adapted system. Replacing the discipline is worth more than tuning it.

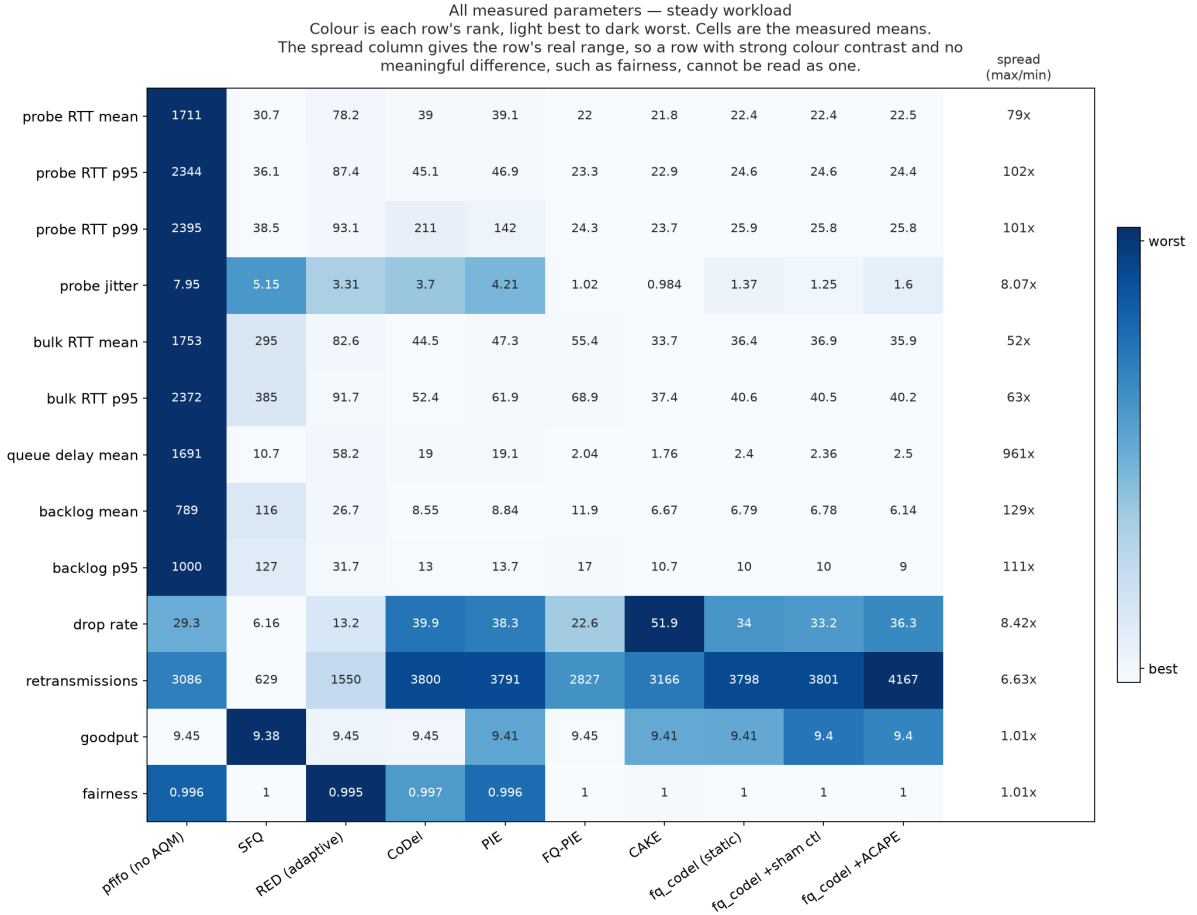


Fig. 2 Every measured quantity against every system under the steady workload. Colour encodes each row’s rank, light best to dark worst, and cells carry the measured means. The spread column gives each row’s true range as the ratio of its largest value to its smallest, so that rows with strong colour contrast but no meaningful difference are not mistaken for findings.

2.3 A scaling law in target/RTT

The results above are taken at a single path RTT. Sweeping the base RTT from 2 to 200 ms while holding the configured `target` fixed shows that the benefit of adaptation is not a function of path RTT at all, but of the dimensionless ratio $r = \text{target}/\text{RTT}$. Fitting the reduction in bulk-flow RTT, b , over 10 cells gives

$$b(r) = \frac{B}{1 + (r_0/r)^k}, \quad B = 20.5\%, \quad r_0 = 0.50, \quad k = 2.6, \quad (1)$$

with $R^2 = 0.986$, where B is the ceiling, r_0 the ratio at which half of it is reached and k the sharpness of the transition. Figure 3 shows the fit, and shows the same measurements plotted against path RTT alone, where they do not collapse.

Hypotheses and decision thresholds were committed to a public repository before any of these data were collected, and three of the four held. A crossover exists: benefit remains under 1.5% and is not statistically significant below $r = 0.25$, and reaches 16.4 to 20.1% at $r \geq 1$. The curve is invariant to path RTT at fixed ratio: cells at 5, 20 and 80 ms with $r = 1$ give 18.4, 17.6 and 16.4%, a spread of 2.0 percentage points against a threshold of 5 set in advance. Leave-one-out prediction of the interior ratios has a mean absolute error of 1.4 percentage points, worst case 3.5. The functional form also transfers to mechanisms the fit never saw: refitted independently over the cells each mechanism has in common, plain `code1` places r_0 at 0.42 against `fq_code1`'s 0.49, and PIE, which shares nothing with CoDel beyond possessing a delay target, places it at 1.06, with R^2 of 0.978 and 0.992 respectively. The 0.49 here is the same quantity as the 0.50 of Equation 1, refitted on the seven cells common to all three mechanisms rather than the full 10; the small difference is the cost of the smaller fitting set and not a second estimate.

One pre-registered test failed and is reported as such. The spread in r_0 across the three mechanisms is $2.52\times$, beyond the $2\times$ threshold set in advance. It is driven entirely by PIE, the CoDel family agreeing to 0.49 against 0.42, and the three disciplines differ in their `target/interval` ratio as well as in mechanism, so the measurement is confounded. The functional form transfers; the constant does not.

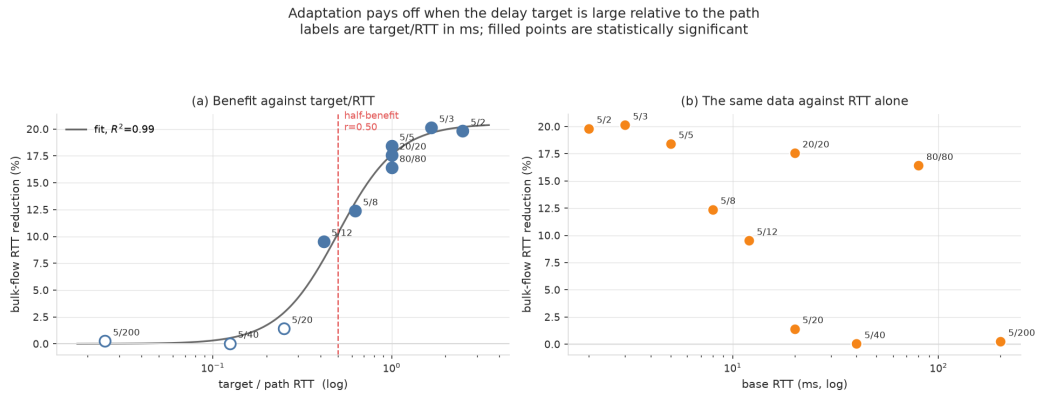


Fig. 3 Benefit of parameter adaptation against the dimensionless ratio $r = \text{target}/\text{RTT}$ (left) and against path RTT alone (right). The same measurements collapse onto a single logistic curve under the ratio and scatter under RTT, which is the evidence that the ratio rather than the delay is the governing variable. Filled points are statistically significant, open points are not.

3 Discussion

Equation 1 has a direct operational reading. Published guidance places a correctly configured deployment at r between 0.05 and 0.10, while half the available benefit arrives only at $r = 0.50$,

five to ten times outside that envelope. This also accounts for the null result of Section 2. Those measurements were taken at the Linux default `target` of 5 ms over a 20 ms path, which is $r = 0.25$, and Equation 1 puts the expected benefit there at 2.9 % against a measured 1.4 %, which three repetitions per cell cannot resolve. The controller was not failing in those measurements; it was being asked to work at a ratio where the law says there is little to recover. On a correctly configured link, therefore, there is almost nothing for parameter adaptation to recover, and the honest description of what such a controller does is that it repairs misconfiguration. The ratio is computable from two numbers an operator already has, the configured target and a path RTT estimate, so the predicate can be evaluated before any controller is deployed.

Where the ratio is small, the measurements point to a different action. Switching from `fq_codel` to CAKE is worth more at every RTT tested than tuning `fq_codel`'s four parameters, and it requires no runtime machinery at all. That the largest available improvement comes from a configuration change rather than from control is not a comfortable result for an adaptive system, but it is what the measurements show.

Several limits bound these claims. A single 10 Mbit/s bottleneck in network namespaces on one kernel is not a datacentre, and the ceiling B should be expected to move with link rate and flow count; the dimensionless form is the portable claim and the fitted constants are not. Three repetitions per cell give wide intervals, which is why the law is fitted across 10 cells rather than rested on any single one. A second dimensionless group, the configured target against the serialisation time of one maximum-size packet, was pre-registered and failed: benefit peaks at intermediate values rather than being floored below unity, so r alone does not close the problem. And the cross-mechanism transfer established the form but not the constant, as described above.

4 Methods

4.1 Testbed

Three Linux network namespaces model a client, a router and a server, connected by virtual Ethernet pairs. The router carries `netem` for base propagation delay and a token-bucket filter at 10 Mbit/s as its root queue discipline, with the discipline under test installed as the child of that shaper on the egress interface facing the server. Placing the bottleneck there puts it on the data path. An earlier two-node arrangement placed it so that it shaped the reverse direction, and every queue measurement taken that way described the acknowledgement path rather than the data path.

4.2 Queue disciplines evaluated

Nine configurations are compared under an identical bottleneck: `pfifo` with no AQM, SFQ, RED, `codel`, PIE, FQ-PIE, CAKE[10], static `fq_codel`, and `fq_codel` under the controller described below. CAKE is configured `besteffort` without its own bandwidth setting so that the token bucket remains the only shaper and every discipline meets the same bottleneck. A tenth arm, the sham controller, polls at the controller's cadence and executes the same classification and prediction code but applies no parameter change; it exists to separate the controller's computational cost from the effect of its decisions.

4.3 Controller

At each control tick the controller reads the queue discipline's own statistics together with per-flow state maintained by an eBPF program attached at the `tc clsact` egress hook[20], classifies a congestion regime from drop rate and backlog, and estimates a trajectory by least squares over a ten-sample window. At each adjustment opportunity it applies an additive-increase multiplicative-decrease rule taken from Adaptive RED[3], with $\beta = 0.9$ on decrease. Observation and adjustment run on separated timescales an order of magnitude apart, following the

two-timescale stochastic approximation framework[21].

Every bound, step size and regime threshold is expressed relative to the configured `target` rather than in absolute milliseconds. Expressed absolutely, a controller started against an 80 ms target would take steps sixteen times smaller in relative terms than one started against 5 ms, and the ratio dependence reported in Equation 1 would then be partly an artefact of the controller’s constants rather than a property of the queue.

Flow-map contents are read through the `bpf(2)` system call directly rather than by invoking an external utility per sample, which reduces the cost of one control tick from 3.17 to 0.709 s and avoids introducing a second process whose scheduling would confound the latency comparison the sham arm exists to isolate.

4.4 Measurements

Three latencies are recorded separately: a sparse ICMP probe for the latency a thin flow experiences, the smoothed round-trip time of the bulk TCP flows as reported by `iperf3`, and a queue-delay proxy computed from instantaneous backlog and the configured rate. They are not interchangeable, and the ratio between the sparse and the bulk latency ranges from 1.0 to 9.6 across the systems tested. Goodput is taken from `iperf3`’s received byte count and flow fairness is reported as Jain’s index[22].

4.5 Experimental design and statistics

Three workloads are used: a steady bulk load of eight TCP CUBIC[23] flows, a staged load whose flow count varies from two to twenty-four during the run so that the congestion regime changes, and a mixed load adding sparse constant-rate UDP probes alongside bulk traffic. Each cell is repeated three times as an independent run. Differences between conditions are assessed with Welch’s t -test, which does not assume equal group variances. Two-sided p -values are computed exactly from Student’s t distribution through the regularised incomplete beta function rather than by normal approximation, which at three samples per group materially overstates significance.

4.6 Platform

Experiments ran on Linux 6.12.48, configured with every evaluated queue discipline compiled in and BPF Type Format enabled, under hardware-assisted virtualisation.

Data availability

All code, kernel configuration, raw measurement logs and analysis scripts are available at <https://github.com/Amritha902/ccn-linux-qdisc-study>. Every table and every numerical value quoted in this article is generated directly from the committed logs by the analysis scripts; none is entered by hand. The pre-registered hypotheses and their decision thresholds are committed to the same repository with timestamps preceding the corresponding measurement runs.

Declarations

Funding. No funding was received for conducting this study.

Conflict of interest. The authors declare no competing interests.

Ethics approval and consent to participate. Not applicable.

Consent for publication. Not applicable.

Materials availability. Not applicable.

Code availability. All analysis and experiment code is available at the repository given above.

Author contribution. A.S. designed and implemented the testbed, the controller and the

analysis pipeline, and wrote the manuscript. Y.P. contributed to the experimental design and the measurement campaigns. D.A. supervised the work. All authors reviewed the manuscript.

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